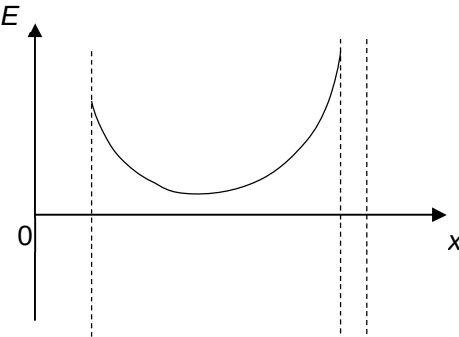


	time taken is longer.
4	
(a)(i) (a)(ii) (a)(iii)	
	Correct general shape for X and Y
	Show an understanding that the resultant potential is a scalar sum of the potential due to the individual charges.

(b)	
	U shaped graph
	Show an understanding that the resultant field strength is the negative gradient of the potential graph.
(c)	For charge to be stationary, net force = 0 $ E_{+q} = E_{-2q} $
	Let the distance of the point from +q charge be r. $\frac{q}{4\pi\epsilon_0 r^2} = \frac{2q}{4\pi\epsilon_0 (d+r)^2}$
	$r = 2.41 d$ (3sf)